HEALTHCARE NURSING STAFFING ANALYTICS

AWS Data Engineering Project — Final Report

Reporting Period: Q2 2024 (April–June 2024)
Platform: Amazon Web Services (AWS)
Project Status: Data pipeline and dashboard completed; GitHub submission preparation in progress
Presentation Format: Live presentation

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1. Executive Summary

This project demonstrates the design and implementation of a cloud-based healthcare data engineering solution using Amazon Web Services (AWS).

The objective was to build a data pipeline that ingests healthcare staffing information, processes and organizes the data, performs analytical transformations, and presents meaningful insights through an interactive dashboard.

The project uses publicly available Centers for Medicare & Medicaid Services (CMS) data concerning nursing homes and skilled nursing facilities.

The solution follows the **Medallion Architecture**, consisting of Bronze, Silver, and Gold data layers.

The completed pipeline supports analysis of nursing staffing hours, staffing patterns across states, and facility-level staffing performance.

The project also examines the four business questions specified in Step 7 of the assignment. Where the available data does not support a definitive answer, the limitations and additional data requirements are documented.

2. Project Objectives

The objectives of this project were to:

1. Establish a cloud-based healthcare data lake using Amazon S3.
2. Ingest raw healthcare staffing and provider datasets.
3. Implement Bronze, Silver, and Gold data layers.
4. Transform raw staffing records into analytics-ready datasets.
5. Use SQL to calculate healthcare staffing metrics.
6. Validate data completeness, uniqueness, and accuracy.
7. Develop an interactive Amazon QuickSight dashboard.
8. Investigate the four healthcare business questions required by the SME.
9. Document the architecture, transformations, and analytical findings.
10. Publish the project materials to GitHub and deliver a live presentation.

3. AWS Technologies Used

Technology	Purpose
Amazon S3	Store raw, processed, and curated healthcare datasets
AWS Glue	Catalog data and perform ETL processing
AWS Glue Data Catalog	Maintain database and table metadata
Amazon Athena	Query and transform healthcare datasets using SQL
Apache Parquet	Store processed data in an efficient columnar format
Amazon QuickSight	Create interactive dashboards and healthcare visualizations
GitHub	Publish project documentation, SQL scripts, and supporting materials

Why These Technologies Were Selected

Amazon S3: Provides scalable object storage for the healthcare data lake.

AWS Glue: Supports automated data discovery, cataloging, and ETL processing.

Amazon Athena: Enables serverless SQL analysis without maintaining a traditional database server.

Apache Parquet: Provides columnar storage that can improve analytical query efficiency.

Amazon QuickSight: Enables interactive visualization of facility-level and state-level healthcare metrics.

GitHub: Provides version control and a central location for project documentation, scripts, and presentation materials.

4. Data Architecture

The project implements Medallion Architecture to organize healthcare data into three layers.

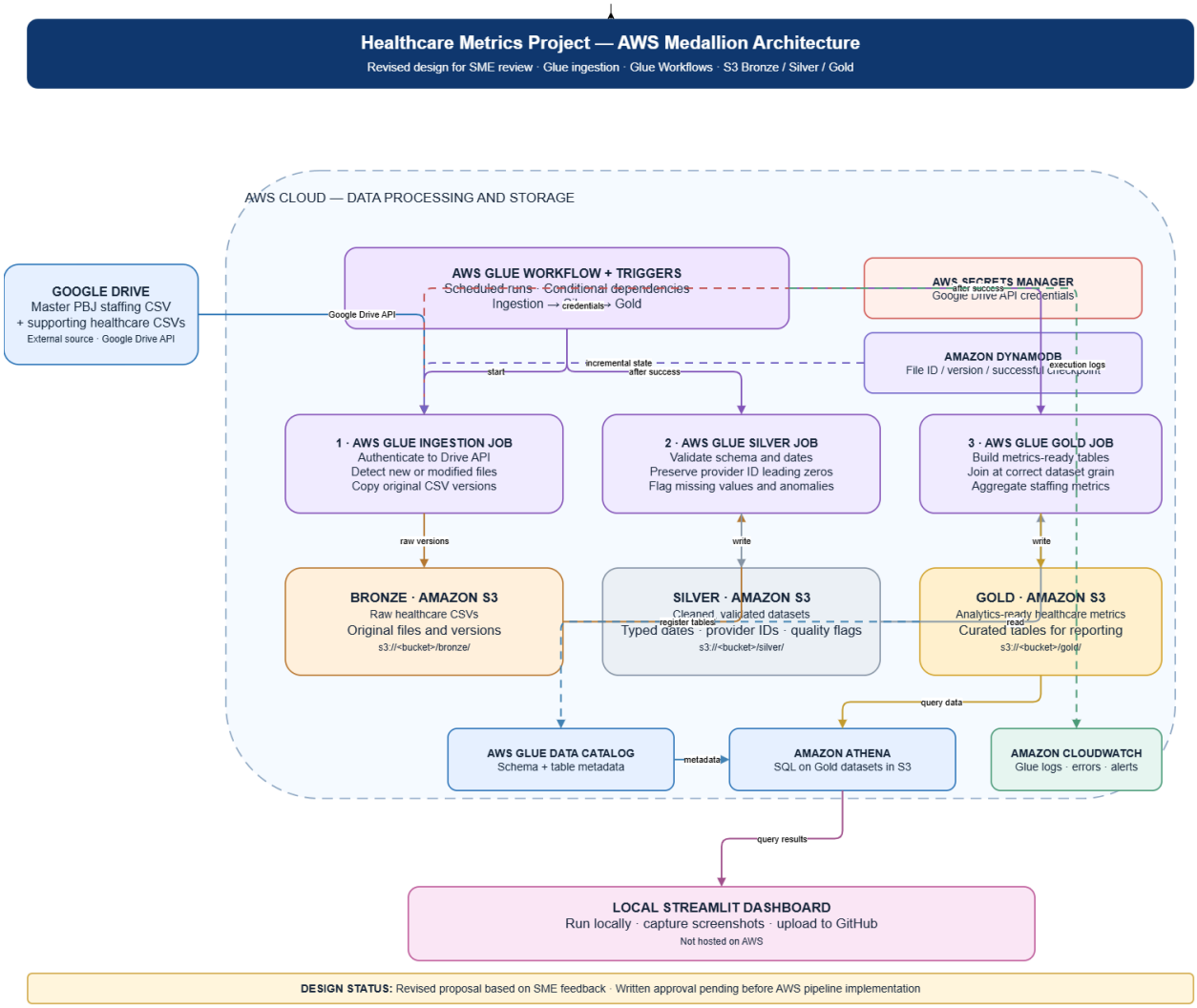


Figure 1: Healthcare Metrics AWS Medallion Architecture

The diagram illustrates the end-to-end healthcare data pipeline using Amazon S3, AWS Glue, Amazon Athena, and Amazon QuickSight, organized into Bronze, Silver, and Gold data layers.

Data Pipeline:

CMS Healthcare CSV Files
↓
Amazon S3 — Bronze Layer
↓
AWS Glue — ETL Processing
↓
Amazon S3 — Silver Layer (Parquet)
↓
Amazon Athena — SQL Transformations
↓
Amazon S3 — Gold Layer (Parquet)
↓
Amazon QuickSight — Healthcare Analytics Dashboard

4.1 Bronze Layer — Raw Data

The Bronze layer stores the original healthcare CSV files.

The data includes nurse staffing records, provider information, and other CMS healthcare datasets.

Purpose:

- Preserve original source records.
- Support auditing and troubleshooting.
- Maintain a source for downstream processing.

S3 bucket:

`healthcare-metrics-datalake-593271843946`

Storage location:

`s3://healthcare-metrics-datalake-593271843946/bronze/`

4.2 Silver Layer — Processed Data

The Silver layer contains processed staffing records stored in Apache Parquet format.

AWS Glue was used to process the source data and register the resulting dataset.

Athena table:

`healthcare_metrics_db.silver_silver`

Storage location:

`s3://healthcare-metrics-datalake-593271843946/silver/`

Purpose:

- Prepare staffing records for analysis.
- Support SQL queries.
- Provide structured data for downstream aggregations.

4.3 Gold Layer — Analytics-Ready Data

The Gold layer contains aggregated facility-level staffing measures.

Amazon Athena SQL transformations were used to create the analytics table.

Athena table:

`healthcare_metrics_db.gold_facility_staffing`

Storage location:

`s3://healthcare-metrics-datalake-593271843946/gold/facility_staffing/`

The Gold dataset includes:

- Facility certification number
- Facility name
- State
- Number of reporting days
- Total registered nurse (RN) hours
- Total licensed practical nurse (LPN) hours
- Total certified nursing assistant (CNA) hours
- Total nursing hours
- Nursing hours per resident day

This dataset serves as the primary source for the Amazon QuickSight dashboard.

5. Data Quality Validation

Data quality checks were performed using Amazon Athena.

5.1 Silver Layer Validation

Metric	Result
Total staffing records	1,325,324
Unique facilities	14,564
Unique reporting dates	91

These results establish the number of staffing records, facilities, and dates represented in the processed dataset.

5.2 Gold Layer Validation

Metric	Result
Total facilities	14,564
Unique facility identifiers	14,564
Missing nursing-hour totals	0
Missing staffing ratios	0
Minimum reporting days	91
Maximum reporting days	91
Invalid reporting-day counts	0
Negative nursing-hour totals	0

5.3 Data Quality Findings

The Gold dataset passed the recorded checks for facility identifier uniqueness, missing staffing measures, reporting-day validity, and negative nursing-hour totals.

All 14,564 facilities in the Gold dataset have 91 reporting days.

These results support the reliability of the aggregated staffing measures for the project's reporting period.

6. Amazon QuickSight Dashboard

Dashboard Name: Healthcare Nursing Staffing Dashboard — Q2 2024

Dashboard URL:

https://us-east-1.quicksight.aws.amazon.com/sn/account/Myriam29/dashboards/dad6af2e-08a9-4d54-8aa8-5f7f000630e4

Dashboard Features

The dashboard includes:

1. Total healthcare facilities.
2. Total nursing staffing hours.
3. State-level nursing-hours comparisons.
4. RN, LPN, and CNA staffing-hour breakdowns.
5. Facility staffing rankings.
6. Interactive state filters.
7. Interactive facility-name filters.
8. Facility-level staffing analysis.

Dashboard Results

Total Facilities: Approximately 14,560

Total Nursing Hours: Approximately 364.15 million

The dashboard provides a visual overview of nursing staffing activity across the facilities represented in the dataset.

Dashboard sharing and access were previously tested by the project owner.

7. Required Business Questions and Analytical Findings

This section addresses the four questions specified in Step 7 of the SME assignment.

Important Data Scope: The project primarily analyzes CMS nursing-home and skilled-nursing-facility records. These datasets are not equivalent to general hospital staffing, hospital payroll, or patient admissions datasets.

Question 1: What is the relationship between nurse staffing levels and hospital occupancy rates?

Analytical Approach

An exploratory occupancy measure was calculated for nursing facilities using average daily resident census and certified-bed counts.

Occupancy proxy:

Average Daily Resident Census / Number of Certified Beds

This occupancy proxy was compared with nursing hours per resident day.

Findings

The analysis included approximately 14,518 usable matched facilities.

The calculated Pearson correlation was approximately **+0.015**.

This indicates virtually no linear association between the occupancy proxy and nursing hours per resident day in the analyzed sample.

Interpretation

The findings do not establish a meaningful linear relationship between the two measures.

However, this analysis does not demonstrate that staffing and occupancy are unrelated in every healthcare setting.

It also does not establish causality.

Limitations

The analysis uses nursing-facility data rather than general hospital occupancy records.

Certified-bed counts came from a later provider-information snapshot, so the occupancy measure is exploratory.

Conclusion

Within the available nursing-facility sample, the analysis found virtually no linear association between the occupancy proxy and nursing hours per resident day.

A definitive hospital occupancy analysis would require hospital-specific occupancy and staffing records from matching reporting periods.

Question 2: Which hospitals have the highest overtime hours for nurses?

Analytical Approach

The staffing dataset was reviewed to determine whether overtime hours could be identified separately from regular nursing hours.

Findings

The current Payroll-Based Journal (PBJ) staffing dataset includes RN, LPN, and CNA staffing hours.

However, it does not contain a separately identified nurse overtime-hours measure.

Interpretation

Total nursing hours cannot be assumed to represent overtime.

Similarly, contract staffing hours should not automatically be classified as overtime.

Additional Data Required

A reliable overtime analysis would require payroll or timekeeping records containing:

- Facility or hospital identifier
- Nurse identifier or staffing category
- Regular hours
- Overtime hours
- Reporting period

Conclusion

The hospitals with the highest nurse overtime hours cannot be reliably identified using the available dataset.

This question requires additional hospital payroll or timekeeping information.

Question 3: What are the average staffing levels by state and hospital type?

Analytical Approach

The available data supports analysis of nursing staffing hours per resident day across states.

It also includes CMS provider classifications.

However, CMS nursing-facility classifications should not be treated as equivalent to general hospital types.

Findings

An exploratory analysis of usable matched nursing facilities produced the following unweighted state or jurisdiction averages:

State/Jurisdiction	Average Nursing Hours per Resident Day
Alaska	Approximately 6.05
District of Columbia	Approximately 4.53
Puerto Rico	Approximately 4.39
Oregon	Approximately 4.20
Hawaii	Approximately 4.12

These figures are exploratory facility-level averages, not validated hospital-type rankings.

Interpretation

Average nursing staffing levels differ across states and jurisdictions.

Such differences may reflect facility characteristics, resident care needs, staffing models, and other factors.

The analysis does not establish which of these factors causes the observed differences.

Additional Data Required

To answer the question specifically by hospital type, the dataset would need a reliable hospital-type field, such as:

- General acute-care hospital
- Critical-access hospital
- Specialty hospital
- Rehabilitation hospital
- Psychiatric hospital

Conclusion

State-level nursing-facility staffing comparisons are supported by the available data.

A complete comparison by state and hospital type requires additional hospital classification and staffing information.

Question 4: What trends can you identify in patient length of stay over time?

Analytical Approach

Patient length of stay is generally calculated using admission and discharge information.

Length of Stay = Discharge Date − Admission Date

To identify trends, average length of stay would be calculated across multiple reporting periods.

Findings

The current staffing dataset includes daily resident census information but does not provide the patient-level admission and discharge records necessary to calculate length of stay.

Interpretation

Daily resident census can indicate how many residents were present on a reporting day.

It cannot independently determine how long each patient remained in a facility.

Additional Data Required

A reliable length-of-stay analysis would require:

- Patient or encounter identifier
- Facility identifier
- Admission date
- Discharge date
- Reporting period

Conclusion

Patient length-of-stay trends cannot be reliably determined from the available staffing dataset.

Additional admission and discharge data covering multiple periods would be required.

8. Provider Data Integration and Limitations

The project also examined integration between the staffing dataset and the CMS provider-information dataset.

Provider Data Results

Metric	Result
Unique provider identifiers	14,814
Gold staffing facilities	14,564
Matched staffing facilities	14,547
Unmatched staffing facilities	17
Provider records with missing certified-bed counts	994

Provider identifiers required normalization before matching.

The analysis showed that 17 staffing facilities did not match the provider-information dataset.

Missing certified-bed counts also limited occupancy calculations.

Consequently, the occupancy analysis was performed using a smaller usable sample rather than every facility in the Gold dataset.

These limitations are important when interpreting the analytical findings.

9. GitHub Repository Deliverables

The GitHub repository will provide documentation and supporting materials for the completed AWS data engineering project.

Planned Repository Contents

1. **README.md:** Project overview, objectives, architecture, technologies, and execution summary.
2. **Architecture documentation:** Description of the Bronze, Silver, and Gold layers.
3. **AWS Glue ETL script:** The actual deployed transformation script, once exported from AWS.
4. **Athena SQL scripts:** Queries used for data transformations and validation.
5. **Data quality results:** Evidence of record counts, uniqueness checks, and other validation results.
6. **Step 7 findings:** Answers to the four required business questions.
7. **Dashboard documentation:** Screenshots, features, metrics, and dashboard URL.
8. **Final project report:** PDF version of this document.
9. **Presentation materials:** Slides and speaker notes for the live presentation.

The repository should not contain AWS access keys, passwords, confidential information, or unnecessary raw healthcare datasets.

10. Live Project Presentation

The project will be presented live rather than through a recorded video.

Proposed Presentation Structure

Slide 1: Project Introduction

Introduce the Healthcare Nursing Staffing Analytics project and its objectives.

Slide 2: Business Problem

Explain the importance of healthcare staffing data and the need for accessible, reliable staffing analytics.

Slide 3: AWS Architecture

Present the pipeline from CMS source data through Amazon S3, AWS Glue, Amazon Athena, and Amazon QuickSight.

Slide 4: Bronze, Silver, and Gold Layers

Explain the purpose of each layer and how the data moves through the pipeline.

Slide 5: Data Transformations and Quality Checks

Describe the SQL aggregations and validation results.

Slide 6: Healthcare Dashboard Demonstration

Show the QuickSight dashboard, KPIs, state comparisons, staffing breakdowns, and interactive filters.

Slide 7: Step 7 Business Questions

Present the findings and limitations for all four required questions.

Slide 8: Technology Decisions and Lessons Learned

Explain why the selected AWS services were appropriate and describe the data engineering challenges encountered.

Slide 9: GitHub Repository

Demonstrate the project documentation, SQL scripts, and other repository materials.

Slide 10: Conclusion and Questions

Summarize the project's accomplishments and answer questions from the SME or audience.

11. Project Outcomes

The project successfully established a cloud-based nursing staffing analytics pipeline using AWS.

The main accomplishments include:

- Building a healthcare data lake in Amazon S3.
- Organizing data using Medallion Architecture.
- Processing staffing data using AWS Glue.
- Creating analytics-ready facility-level measures in Amazon Athena.
- Validating more than 1.3 million daily staffing records.
- Producing a Gold dataset containing 14,564 unique facilities.
- Developing an interactive Amazon QuickSight dashboard.
- Investigating the four required healthcare business questions.
- Documenting analytical limitations and additional data requirements.

The project demonstrates practical experience with data ingestion, cloud storage, ETL processing, SQL transformation, data validation, data modeling, and business intelligence.

12. Final Conclusion

The Healthcare Nursing Staffing Analytics project demonstrates an end-to-end AWS data engineering solution that transforms raw healthcare staffing records into actionable analytical information.

Through the implementation of Bronze, Silver, and Gold layers, the project establishes a structured approach to healthcare data processing.

The Amazon QuickSight dashboard provides an accessible interface for examining staffing measures across facilities and states.

The project also demonstrates the importance of data quality, reliable dataset integration, and accurate interpretation of business questions.

Although some of the SME's requested analyses require additional hospital-specific or patient-level data, the project identifies these limitations and explains what would be needed to complete those analyses reliably.

The completed pipeline and dashboard, together with the final documentation, GitHub repository, and live presentation, provide a practical demonstration of AWS data engineering and healthcare analytics skills.

End of Healthcare Metrics Final Project Report

Dashboard: Healthcare Nursing Staffing Dashboard — Q2 2024
Cloud Platform: AWS
Architecture: Bronze / Silver / Gold
Presentation: Live
GitHub Submission: Pending repository preparation and upload